Carrying Immersion over Networks: A Survey of Networked Extended Reality

ANONYMOUS, To be completed, To be completed

Extended reality (XR) puts the network inside a perceptual loop. When the user's head turns, the new view must appear within a few tens of milliseconds. Once content, computation or other participants are remote, the network must carry 360-degree and volumetric media at high rates, return rendered frames inside that deadline, take a continuous uplink of pose and sensor data, and keep several people and machines consistent. This survey organises networked XR by the six functions that every such system performs. It must *encode* the scene into something that can be sent, *anticipate* what the user will need next, decide where to *compute*, *provision* the network, *connect* people and machines that share a world, and *assure* that the experience is good enough and safe. For each function we state the design question and group the approaches into families by their underlying idea. We then separate what has been established on real systems from what is contested and what is merely assumed. Two couplings tie the functions together: a shared deadline that every function spends from, and a trade between network and device that every design shifts one way or the other. Across all six functions the same weaknesses recur. Objectives are rarely validated against what users perceive, requirement figures are inherited rather than measured, and controllers are seldom tested beyond the simulator. The uplink, multi-user scale and the latency cost of security are studied least. We close with nine cross-cutting gaps and a dependency-ordered research agenda.

CCS Concepts: • **Networks** → **Mobile networks**; *Network performance evaluation*; • **Human-centered computing** → **Virtual reality**; *Mixed / augmented reality*; • **Information systems** → *Multimedia information systems.*

Additional Key Words and Phrases: extended reality, virtual reality, augmented reality, 360-degree video, volumetric video, viewport prediction, edge computing, remote rendering, split rendering, motion-to-photon latency, 5G, 6G, tactile internet, holographic communication, quality of experience

ACM Reference Format:
Anonymous. 2026. Carrying Immersion over Networks: A Survey of Networked Extended Reality. *ACM Comput. Surv.* 1, 1 (October 2026), 41 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

A head-mounted display shows a world that must respond to the head that wears it. When the user turns, the new view must appear within a few tens of milliseconds, or the image lags the body and comfort fails [36]. On a standalone device this deadline is met locally. Once the scene, the computation or the other participants are elsewhere, the network becomes part of the perceptual loop.

Extended reality (XR), the umbrella term for virtual, augmented and mixed reality, therefore asks more of a network than video or the web, in four ways. The first is *rate*. A 360-degree or volumetric scene sent whole needs hundreds of megabits per second, although the user sees only a fraction of it at any moment [90, 161]. The second is *deadline*. Remote rendering must return a frame within the motion-to-photon (MTP) budget, so delay is not merely annoying but perceptually disqualifying [36]. The third is *direction*. The device sends pose, camera frames and captured media upstream all the time, so XR is not a download. The fourth is *sharing*. When several people inhabit one world, or one person operates a remote machine, the network must keep endpoints consistent with each other, not just deliver media to each [13].

Author's address: Anonymous, To be completed, , To be completed.

2026. ACM 0360-0300/2026/10-ART
https://doi.org/10.1145/nnnnnnn.nnnnnnn

Table 1. Representative surveys of networked XR and the functions of Section 3 that each treats as a primary subject (●) or in part (○). Earlier surveys follow one medium or one technology; this survey covers all six functions and the couplings between them.

Survey	Year	Scope	Enc.	Ant.	Comp.	Prov.	Conn.	Assure
Bastug et al. [13]	2017	Interconnected VR		○	○	●	○	
Elbamby et al. [36]	2018	Low-latency, reliable VR		○	●	●		
Antonakoglou et al. [7]	2018	Haptic communication	●			○	●	○
Sukhmani et al. [136]	2018	Edge caching and computing		●	●	○		
Fan et al. [39]	2019	360-degree video streaming	○	●		○		○
Yaqoob et al. [161]	2020	Adaptive 360-degree streaming	○	●				○
Promwongsa et al. [111]	2020	Tactile internet	○			●	●	
Liu et al. [90]	2021	Point-cloud streaming	●	●				○
Siriwardhana et al. [128]	2021	Mobile AR with edge computing			●	○		
Xu et al. [157]	2022	Edge-enabled metaverse			●	●	○	○
Hazarika and Rahmati [49]	2023	Ultra-low-latency XR			○	●		○
Ali et al. [5]	2023	Metaverse networking and security			○	●	○	●
This survey	2026	Networked XR as a whole	●	●	●	●	●	●

Research on these demands is scattered across communities that seldom read each other. Multimedia-systems researchers built viewport-adaptive streaming [39, 161]. Mobile-computing researchers offloaded AR recognition to the edge [128]. Wireless researchers framed XR as a joint allocation of radio, computing and energy. Haptics researchers pursued the tactile internet [7], and 6G and metaverse visions placed all of it inside a future network [5, 157]. Existing surveys follow these boundaries. Each covers one medium, such as 360-degree video [39, 161, 174], point clouds [90] or haptics [111], or one technology, such as edge computing [128, 136], ultra-low latency [49] or the 6G metaverse [123, 157]. What is missing is a single frame that shows how these parts fit together, where they trade against each other, and how far each has been tested. Table 1 places representative surveys against the six functions used in this survey.

Organising idea. We organise the field by *function*: the six things any networked XR system must do, whatever its medium and access network (Section 3). A system must *encode* the scene into something that can be sent (Section 4). It must *anticipate* what the user will look at or do next, so that the right content arrives in time (Section 5). It must decide where to *compute*, on the device, at the edge or in the cloud, and how to hide the delay this adds (Section 6). It must *provision* the network with links, schedulers, transport and architecture that supply the rate, delay and reliability required (Section 7). It must *connect* users with each other and with remote machines, through shared state, touch and telepresence (Section 8). And it must *assure* that the result is good enough and safe for the people who use it (Section 9). Each function poses its own design question. Within each, we group approaches into a few families by their underlying idea, so that methods described in different vocabularies can be compared on the same terms.

Contributions.

(1) A **functional framework** that places the problems of networked XR on one reference architecture and defines each function by a design question (Section 3).
(2) For each function, a **classification of approaches into families**, stating what each family assumes, when it wins and where its limits lie, with the measured results that support each claim (Sections 4 to 9).
(3) An account of the **couplings between functions**: the shared deadline and the trades between network and device that tie them together.
(4) **Nine cross-cutting gaps** and a research agenda ordered by dependency (Sections 10 and 11).

Organisation. Section 2 introduces the XR loop, the motion-to-photon deadline and the properties of XR traffic for readers new to the area. Section 3 presents the functional framework. Sections 4 to 9 treat the six functions in loop order, and each closes with its open problems. Section 10 draws the cross-cutting gaps together, Section 11 proposes a research agenda, and Section 12 concludes.

2 BACKGROUND: THE NETWORKED XR LOOP

This section introduces the concepts that the rest of the survey relies on. Readers familiar with XR systems may skip to Section 3.

2.1 From standalone to networked XR

XR spans a continuum. Virtual reality (VR) replaces the user's view with a synthetic one. Augmented reality (AR) overlays virtual objects on the physical world, and mixed reality anchors them in it so that they respond to physical surfaces. Each device tracks the user's head, and often the hands and eyes, and redraws the scene for every new pose. Two levels of freedom matter for networking. With *three degrees of freedom* (3DoF) the user can only rotate the head, as in a 360-degree video. With *six degrees of freedom* (6DoF) the user can also move through the scene, as in a game or a volumetric capture. A 3DoF scene can be prepared in advance as a sphere of pixels. A 6DoF scene must be rendered or synthesised for each viewpoint, so it needs either geometry on the device or rendering close to it.

A standalone headset performs the whole loop locally: it senses a pose, renders, and displays the result. Three pressures push parts of this loop into the network. First, *content* may be too large to store, or may be produced live elsewhere, as with 360-degree broadcasts and volumetric captures. Second, *computation* may exceed what a light, battery-powered device can sustain, as with photorealistic rendering and AR scene understanding [128]. Third, *other participants*, whether people or machines, are by definition remote [13].

2.2 The motion-to-photon deadline

The defining constraint of XR is the *motion-to-photon* (MTP) latency: the time from a head movement to the photons that show its consequence. When MTP exceeds roughly 15–20 ms the displayed world appears to lag or swim, which degrades presence and can cause simulator sickness. That budget leaves roughly 14 ms for computation and communication together [36]. Three properties of this deadline shape every design in this survey. First, it is *per frame*: at 72–120 frames per second, each frame has its own deadline, and a late frame is a lost frame. Second, it applies to *rotation* most strictly. Local reprojection can correct a slightly stale frame for a new head orientation, so the network-facing deadline for full interaction is looser than MTP, although it is still tight. Third, its *tail* matters more than its mean, because occasional late frames are what users notice.

Table 2. How networked XR traffic differs from conventional video streaming. Each row names a property, its consequence for the network, and the section that treats it.

Property	Video streaming	Networked XR	Treated in
Deadline	Seconds of buffer absorb delay	Each frame must arrive within a few tens of milliseconds	Sections 6 and 7
Direction	Downlink, with small requests upstream	Continuous uplink of pose, input, camera and capture	Sections 6 and 8
Viewport	The whole frame is shown	Only the viewport is seen, and it moves	Section 5
Representation	2D video frames	Spherical video, point clouds, meshes, neural fields, haptic signals	Section 4
Participants	Independent viewers	Shared state, avatars, remote machines	Section 8
Quality	Rebuffering, bitrate, video quality	Latency, judder, sickness, presence, task success	Section 9

2.3 What XR traffic looks like

The traffic of a networked XR session differs from video streaming or web traffic in four ways (Table 2). It is *periodic and bursty*: rendered or decoded frames arrive at the display rate, each a burst of packets with its own deadline. It is *bidirectional*: pose, controller input, camera frames and captured media flow upstream continuously, and the downstream content depends on them. It is *viewport-dependent*: the user sees only a fraction of a 360-degree or volumetric scene at any moment, so the useful content depends on where the user looks. And it is often *shared*: several users' updates must be reconciled into one consistent world. The first two properties make XR a real-time control problem rather than a throughput problem. The third creates the opportunity to send less. The fourth makes the network responsible for consistency as well as delivery.

2.4 Where the loop can run

A networked XR system distributes its loop over four tiers: the *device*, the *access network* (Wi-Fi or cellular), the *edge* (servers at or near the access network) and the *cloud*. Moving a function to a farther tier gains resources and loses time. An edge server one wireless hop away adds a few milliseconds of round-trip time, while a distant cloud data centre adds tens of milliseconds [41]. Mobile edge computing, as standardised for 5G, exists largely to put rendering, caching and session state within the XR deadline [136]. The access network is usually the least predictable part of the path. Wireless capacity, delay and loss vary with distance, blockage, interference and the number of users, and a directional millimetre-wave beam must follow a head that turns quickly. The framework in the next section builds on these four tiers.

3 A FUNCTIONAL FRAMEWORK FOR NETWORKED XR

Networked XR is usually described by its parts: latency, caching, 5G, point clouds, holographic communication. These labels mix levels. Latency is a property, caching a mechanism, 5G a technology and holographic communication an application. Lists of such topics therefore do not show how the parts depend on each other. We instead describe the field by *function*, the job a part of the system performs in the loop between the user's body and what the user perceives. Each function answers one design question that any networked XR system must answer, whatever its medium or access

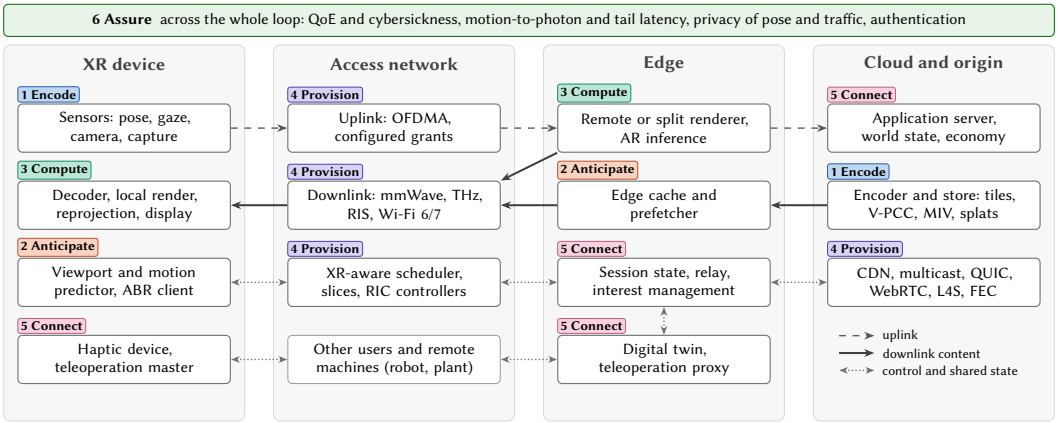


Fig. 1. Reference architecture of a networked XR system, with the six functions placed where they act. Rows read left to right as the uplink of pose and captured media, right to left as the downlink of content, and as control and shared state that couple users and remote machines. The same function often acts at several tiers: anticipation runs both on the device (prediction) and at the edge (caching), and computation is split between device and edge.

network. Figure 1 places the six functions on a reference architecture, and Table 3 summarises the families of approaches that each function contains.

The six functions.

Encode (Section 4). *What representation of the scene travels, and how is it compressed?* Projections and tiling of 360-degree video, point clouds and meshes, multi-view video, neural scene representations, holograms and haptic signals.

Anticipate (Section 5). *What will the user need next, and how is it delivered in time?* Viewport-adaptive streaming, head, gaze and 6DoF motion prediction, and caching and prefetching.

Compute (Section 6). *Where does each stage of perception and rendering run, and how is the added delay budgeted and hidden?* Offloading, remote and split rendering, AR inference at the edge, motion-to-photon budgets, reprojection and device energy.

Provision (Section 7). *How do radio, scheduling, transport and architecture supply the rate, delay and reliability required?* Resource allocation, mmWave, terahertz, reconfigurable surfaces and Wi-Fi, congestion control and loss resilience, and 6G and metaverse architectures.

Connect (Section 8). *How are several people and remote machines kept consistent and responsive?* Multi-user synchronisation, avatars, haptic teleoperation, holographic telepresence and digital twins.

Assure (Section 9). *How do we know the result is good enough, and safe?* QoE measurement and modelling, traffic characterisation, and security, privacy and trust.

Why functions. The grouping has three useful properties. First, each function is *one decision.* Encoding chooses a representation and anticipation a fetch schedule. Computing chooses a split point and provisioning an allocation. Connecting chooses a consistency model and assuring a definition of "good enough". Approaches that answer the same decision can be compared directly, even when they come from different communities. Second, the functions are *ordered along the loop.* Encoding and anticipation decide what is sent, computing and provisioning decide where and how,

and connecting and assuring concern the people at the ends. Third, the functions are *coupled*, and the couplings are where most design difficulty lies.

How the functions couple. Two couplings run through the whole review. The first is the *shared deadline*. A head-worn display tolerates about 15–20 ms from motion to photon, which leaves roughly 14 ms for computation and communication [36]. Every function spends from this one budget. A longer prediction horizon in anticipation relaxes the network but costs accuracy. A remote split in computing adds a round trip that provisioning must keep short and that reprojection must partly hide. A heavier codec in encoding saves rate but adds encoding delay. The second coupling is the *trade between network and device*. Almost every design moves cost from one to the other. A compact representation must be decoded or synthesised on the headset. Rendering at the edge saves the headset's GPU but spends its radio and battery, and prefetching saves delay but spends bandwidth on content that may never be seen. Reading the sections that follow with these two couplings in mind explains why results rarely transfer between settings. A gain obtained under one budget, or one division of cost, is not a gain under another.

4 WHAT IS SENT: REPRESENTING AND ENCODING IMMERSIVE CONTENT

Every networked XR system must first decide what its content is: a sphere of pixels, a set of camera views, a cloud of points, the parameters of a learned scene model or a stream of force samples. The design question is which representation travels and how it is compressed. The choice fixes four quantities that every later function inherits. The *bitrate floor*, below which quality collapses, sets what the network must supply (Section 7). The *encoder delay* counts against the interaction budgets of Sections 6 and 8. The *decoder cost* lands on a battery-powered headset or phone. *Viewport dependence* decides whether the stream can be cut to what the user will see, and so whether prediction (Section 5) has anything to act on.

We group representations into four families by what they send (Table 4). *Image-based* representations send what cameras see. *Geometry-based* representations send the shape of the scene. *Learned* representations send a model of the scene, or its meaning, for the receiver to regenerate. *Signal-based* representations send a physical signal sampled by the rules of the sense that receives it. Figure 2 follows one chain per family from capture to display.

4.1 Image-based: send what a camera sees

Image-based representations keep the 2D video chain and its hardware codecs. Their open questions are how the scene maps onto rectangles and how many decoders the client can afford.

Projected 360-degree video assumes a viewer who turns the head but stays in place (3DoF), and wins when that holds. Concentrating bits on the viewport saves bitrate, but less than it saves pixels: Oculus's offset cube used fewer than half the pixels of an equirectangular frame, yet saved only 5.6–16.4% of bitrate [171]. The delivery unit is contested. Viewport-specific versions multiply storage; tiles share one encode but cost 7.3–10.4% in coding efficiency, and Flare's tile decoders raised phone CPU load from 18% to 58% [113]. Gaze is the stronger lever. Gaze-based quantisation saved 40–47% of bitrate in cloud VR gaming [175], and gaze-contingent compression was five times more effective when motion-to-photon (MTP) latency stayed below 15 ms [54]. Foveation is thus a coding gain that Section 6 pays for in latency.

Multi-view plus depth sends colour and depth streams and lets the receiver synthesise the view, giving limited six degrees of freedom (6DoF) with ordinary video codecs. It assumes the receiver can afford view synthesis, and it wins in live, two-way systems because it needs no slow 3D encoder. Project Starline sends seven H.265 streams at 30–100 Mbit/s and fuses them at the receiver, because

Table 3. The six functions at a glance. Each row names the design question, the families of approaches compared in the corresponding section and the trade-off that all of them negotiate.

Function	Design question	Families of approaches	Central trade-off
Encode Section 4	What representation of the scene travels?	Image-based (projected and tiled video, multi-view plus depth); geometry-based (point clouds, meshes); learned (radiance fields, Gaussian splats, semantic codecs); signal-based (haptic, holographic)	Rate on the wire against encoding delay and decoding cost on the device
Anticipate Section 5	What will the user need next, and how does it arrive in time?	What to predict (motion, viewport, gaze, saliency); what to fetch (tile and quality selection, heuristic, optimised or learned); where to keep it (buffer, edge cache, multicast)	Prediction horizon against accuracy; bandwidth saved against content wasted
Compute Section 6	Where does each stage run, and how is the delay hidden?	Full remote rendering; split and hybrid rendering; perception offload for AR; latency hiding by prediction, reprojection and speculation	Device compute and energy against network delay inside the deadline
Provision Section 7	How does the network supply rate, delay and reliability?	Link (mmWave, THz, RIS, Wi-Fi); scheduler (optimisation, heuristics, learning, XR-aware mechanisms); transport (delay-based and L4S control, FEC); architecture (edge, slicing, 6G)	Per-frame guarantees against shared capacity and efficiency
Connect Section 8	How are people and machines kept consistent and responsive?	Shared state (authoritative, replicated, interest-managed); people (avatars to holoportation); touch (teleoperation control, haptic coding); places (digital twins, freshness)	Consistency and fidelity against responsiveness and scale
Assure Section 9	How do we know it is good enough, and safe?	Experience (user studies, standard, objective and learned QoE models, traffic characterisation); trust (attack surfaces, defences)	Proxy metrics against perceived quality; protection against added delay

real-time 3D compression was impossible at its resolution [75]. MPEG Immersive Video (MIV) standardises this path [78], yet no study has streamed MIV to users over a real network.

4.2 Geometry-based: send the shape

Geometry-based representations send the scene itself, so the receiver can render any viewpoint. The price is weight: raw point clouds need 720 Mbit/s at 200K points and 30 fps [47], one to three orders of magnitude more than 2D video.

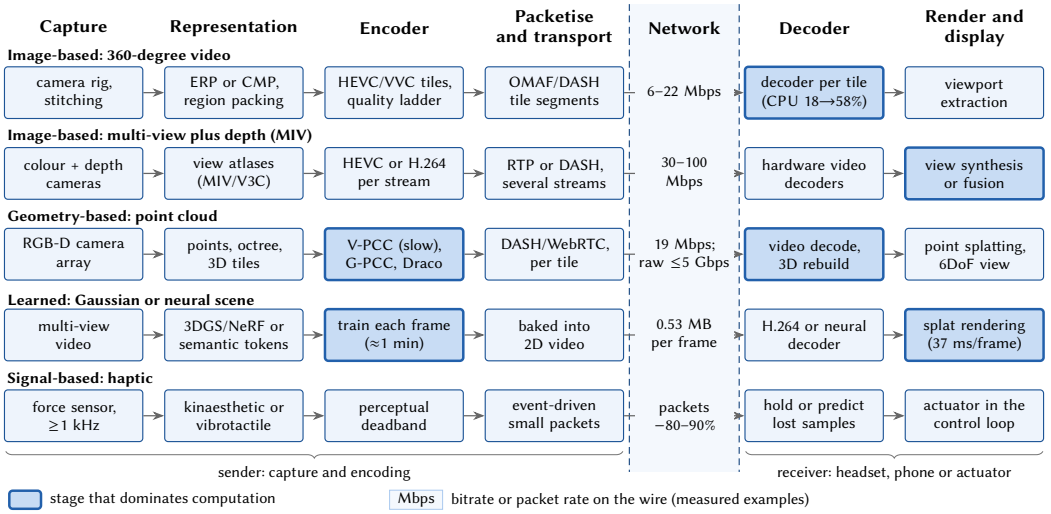


Fig. 2. Transmission chains from capture to display for one or two representations of each family. No chain is cheap at both ends: a low rate on the wire is bought with encoder delay (V-PCC, per-frame training) or with computation on the receiver (tile decoding, view synthesis, splat rendering). Figures: 360-degree video [113, 171], multi-view streams [75], point clouds [47, 144], Gaussian video [145], deadband coding [130].

Point-cloud codecs trade compression ratio against speed, and the trade is established. V-PCC projects patches onto planes and reuses video codecs; it reaches lossy ratios of 100–500 but took about 4 hours to encode 30 frames [141]. G-PCC and Draco code geometry in 3D, run fast and compress less, so live systems avoid V-PCC. In VR2Gather, the MPEG Anchor codec cut 281.8 to 19.0 Mbit/s but raised latency from 40.6 to 278.7 ms, and parallel tiles brought it back to 90.6 ms [144]: coding, not transit, set the delay. The receiver is the second constraint: Hermes reached 30 fps on phones only by exploiting inter-frame correlation [146]. No codec wins everywhere; in a comparison of five codecs the best choice depended on content and viewing behaviour [160].

Geometry's advantage is a richer viewport dependence: points can be culled by visibility, occlusion and distance, not only by direction. ViVo saved 40% of bandwidth at SSIM above 0.99, but only 10% for phone users [47]. Meshes are less explored: a reconstructed mesh for robot teleoperation needed 14 Mbit/s on average [131].

4.3 Learned: send a model or a meaning

Learned representations replace explicit content with something the receiver must regenerate. They promise the largest savings and move the most work to the two ends.

Neural scene representations (NeRF, 3D Gaussian splatting) send model parameters. The most deployable design reuses existing hardware: V^3 bakes dynamic Gaussians into 2D video for H.264 decoders and reaches 32.97 dB PSNR at 0.532 MB per frame and 27 fps on a phone [145]. At 30 fps that is about 128 Mbit/s, the range of multi-view video, so the gain is rendering freedom rather than bitrate. Training takes about a minute per frame, which rules out live content, and error control and quality assessment remain open [124].

Semantic and generative codecs send meaning and assume that sender and receiver share a knowledge base or a generative model. Their comparisons are the weakest of any family. WiserVR sent 1.81 instead of 28.3 Mbit for 20 frames, in an AWGN simulation that ignored the cost of sharing the knowledge base [153]. The one generative prototype timed end to end cut bandwidth by 99.93%

but raised latency from 980 ms to 13.66 s [122]. Fidelity metrics differ between studies and can disagree with perceived realism [164], and baselines are usually BPG with LDPC rather than HEVC or VVC. An order-of-magnitude saving is plausible; its price in inference delay and energy is assumed rather than measured.

4.4 Signal-based: send the physical signal

Signal-based representations code a physical quantity by the limits of the sense that receives it. Haptic and holographic content sit at opposite ends of the rate scale.

Haptic signals need little rate and strict timing. Kinaesthetic samples travel at 1 kHz or more to keep the control loop stable, and perceptual deadband coding sends a sample only when its change exceeds a Weber fraction (7–15% for force), removing 80–90% of packets without perceptible loss. Two independent implementations of the IEEE 1918.1.1 codec reproduced the same rates and errors [130], an established result, though for one contact point. Vibrotactile coding is open-loop and tolerates tens of milliseconds of transform delay [51]; learned tactile codecs report 80–92% cuts, but only against other learned codecs [129]. Deadband coding makes traffic bursty, unlike the periodic 1 ms flow that transport studies assume.

Holographic content has no agreed representation, so its requirement figures depend on an unstated one. A light-field display example needs 221 GB per frame, or 6.63 TBps at 30 fps [3]. Measured systems used 5–100 Mbps for compressed point clouds over live 5G [37] and 173 Mbps for light-field telepresence [15]: one figure is uncompressed display arithmetic, the other compressed capture. For live use, codec delay binds first: V-PCC needs 11–42 minutes to encode one second [57].

The choice moves cost; it does not remove it. No representation is cheap at both ends. Image-based content is light on the sender and heavy on the wire and the decoders. Geometry cuts the wire at the price of encoder delay. Learned representations cut it further and move the work into training and inference. Choosing a representation is choosing where to spend: on the network, the sender's encoder or the receiver's processor. Nor is the choice independent. Remote rendering (Section 6) collapses any representation back into 2D video at the edge, trading decoder cost for MTP latency; viewport dependence bounds what prediction can save (Section 5); encoder delay decides which representations two-way telepresence can use (Section 8). The ratio–speed trade within each family is established; that new representations win across families is assumed, since none has been compared with the others at matched quality.

Open problems

- **P1. Which representation minimises end-to-end cost?** Point clouds, meshes, multi-view video and Gaussian splats have never been compared on the same content at matched perceived quality. Streaming shared scenes over replayed 5G and Wi-Fi traces would measure rate, encoder delay and decoder load.
- **P2. Do semantic and generative savings survive latency?** Reported savings come from AWGN simulations with weak baselines. Comparing semantic codecs with HEVC, VVC and V-PCC at matched quality on a real 5G link, timing inference at both ends, would show whether they suit interactive use.
- **P3. How should new representations be judged?** No rating data exist for compressed Gaussian, neural or generative reconstructions viewed in a headset, so PSNR and LPIPS are unvalidated for their artefacts. A 6DoF rating study across codecs and rates would provide ground truth.

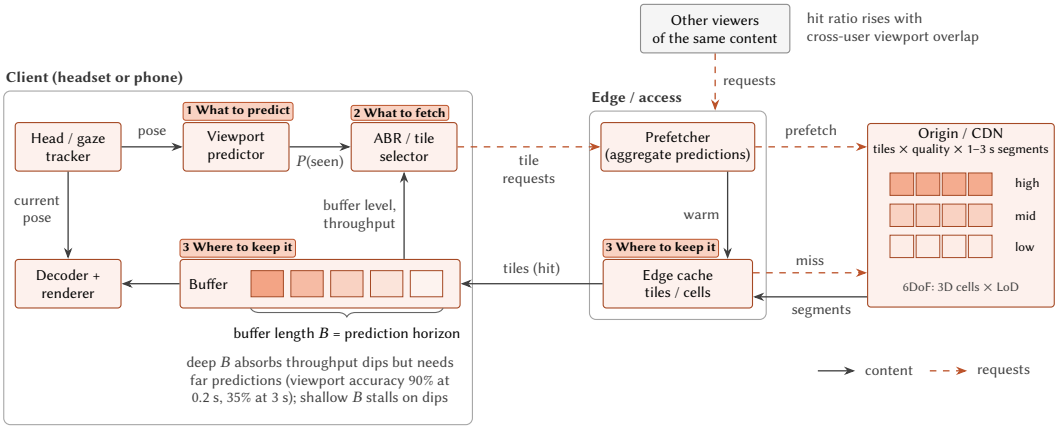


Fig. 3. Viewport-adaptive delivery of tiled 360-degree or volumetric content across client, edge and origin, with the three anticipation decisions marked. Each decision is a bet on where the user will look: the buffer length fixes how far ahead the predictor must see, and the edge cache pays off only when viewers' viewports overlap.

P4. Can codecs track a mobile link? Rate control is tuned at fixed rates, and neural codecs lack bitrate ladders and loss handling. Rate control driven by measured capacity, judged by frame-deadline misses, would test whether codec gains hold on cellular links.

P5. What does a hologram actually cost to send? Requirement tables rest on display arithmetic, not captured content. Coding one scene as point clouds, splats, light fields and phase holograms and recording the traffic would replace assumed figures with measured ones.

P6. How do coded haptic streams behave under real loss? Haptic codecs are evaluated on one-off recordings with one contact point. Shared multi-point recordings and a user study of deadband and vibrotactile codecs over replayed impairments would test perceptual transparency.

5 DELIVERING AHEAD OF THE USER: PREDICTION, ADAPTATION AND CACHING

An immersive client displays only a fraction of what it could receive, and it must request that fraction before the user looks there. Delivery is therefore a bet on the user's future attention. Three decisions place the bet: *what to predict* about the user, *what to fetch* given that prediction and the network, and *where to keep* content close to the user. They form one loop (Figure 3) in which the prediction horizon, the buffer depth and the cost of a wrong commitment constrain one another; Table 5 classifies the approaches. Tiling belongs to Section 4, rendering delay to Section 6, and transport to Section 7.

5.1 What to predict: horizon against accuracy

Prediction trades horizon for accuracy. The target is head orientation for 3DoF video and the full 6DoF pose for volumetric content, where position and occlusion change what is visible. Accuracy falls steeply with horizon. On a 130-user trace set, Flare's best predictor reached median accuracies of 90.5%, 72.9%, 58.2% and 35.2% at 0.2, 0.5, 1 and 3 s [113], and a cross-dataset benchmark found no algorithm accurate much beyond 300 ms [1], whereas streaming buffers hold seconds of content.

Three families of predictors address this gap. *Extrapolators* (persistence, linear regression, Kalman filters) assume smooth motion over the horizon and need no training. *Learned sequence models* (LSTM, GRU, transformer) learn from the user's past trajectory. *Content-aware and cross-user*

predictors add what a trajectory cannot know: saliency [38], gaze, or earlier viewers' paths, the strongest long-horizon signal for on-demand content. Cross-user prediction raised accuracy by 20.2 percentage points [10], and other users' paths helped most at 4–10 s [79]. Live and interactive content lack this history, so these results do not transfer to them.

Whether learned models beat simple ones is contested. On the Tsinghua dataset, persistence reached a hit rate of 0.8355, against 0.8402 for sequence-to-sequence and 0.8464 for LSTM models [79]; for 6DoF pose, simple statistical models performed comparably [58]. Learned models report large gains mostly over each other: a fusion transformer's average precision of 0.954 at 1 s, against 0.901 for the previous best, came from a random split that shared videos and users between training and test [168]. Without extrapolation baselines and video- and user-disjoint splits, accuracy cannot be separated from leakage. A second assumption is that a point estimate suffices, although the fetcher needs a distribution to size its margin; predicting the predictor's own deviation saved 45% of bandwidth at under 0.1% failure [11], yet calibration of predicted probabilities is rarely reported.

5.2 What to fetch: turning probabilities into requests

The fetcher turns a prediction and a throughput estimate into requests for spatial units, qualities and segments. Its objective, a hand-weighted sum of viewport quality, rebuffering and switches, is assumed rather than measured. Hedging against prediction error is the common core: 360ProbDASH fetched tiles in proportion to their viewing probability and raised viewport PSNR by about 39% [155], and most designs add low-quality margin tiles.

The controllers fall into three families. *Heuristic, buffer-based controllers* map buffer level and view probability to quality with fixed rules; BOLA360 adds a near-optimality proof without training and reports 6–110% QoE gains [163]. *Optimisation-based controllers* solve a rate-utility allocation, often a knapsack over tiles or 3D cells [107]; they are exact for their model and fragile when its inputs are wrong. *Learned controllers*, mostly deep reinforcement learning, train on trace replay and report 20–30% gains over heuristics [167], but degrade on unseen viewing patterns and networks [152]. No study has compared buffer-based and learned controllers on shared traces under one objective.

Six degrees of freedom change the unit of adaptation from a tile to a 3D cell, an object or a level of detail (LoD). ViVo combined viewport, distance and occlusion visibility and saved 40% of data on average at SSIM above 0.99 [47]. No general allocation policy has emerged: the best multi-object strategy depends on the scene and the camera path [142].

Live delivery removes the pre-encoded quality ladder and the cross-user history, and two-way volumetric communication breaks the buffer model altogether: a tiled user-adaptive point-cloud system needed 14 Mbps instead of about 300 Mbps in a 33-user study [135]. In such systems adaptation is coupled loosely, if at all, to congestion control (Section 7).

5.3 Where to keep it: buffers, caches and shared delivery

Keeping content close buys the right to be wrong cheaply, and it pays most when viewers' attention overlaps. *Popularity caches* (LRU, LFU) treat tiles as independent objects and are what deployed HTTP caches run. *Viewport-aware caches and prefetchers* place tiles by cross-user viewing: viewport-similarity placement raised the edge hit ratio by at least 40% over LFU and 17% over LRU [93], and after 24 views OpCASH served over 95% of the viewport from cache while fetching 85% less from the origin [92]. Independent work replicates the gain [17], but only against popularity baselines, never an oracle such as Belady's. *Shared delivery* exploits overlap on the air rather than in storage: multicast of similar viewports saved over 50% [12], and coded caching promises gains that grow with the number of users, although subpacketisation and synchronised requests may erase them [121].

Three questions are open. Cache hits distort the throughput estimates of DASH clients [94], yet most caching designs assume an ideal client. Whether learned policies add value cannot be settled while XRFlux remains the only shared benchmark [4]. That hit ratio stands for experience is assumed: no XR caching scheme has been evaluated with viewers. Caching of 6DoF content remains almost unstudied.

5.4 One loop: horizon, misprediction and realistic savings

The three decisions share one variable, the time between commitment and display. A deep buffer absorbs throughput dips but commits bits on predictions seconds ahead; a shallow one needs short predictions but stalls on dips. Buffer length is therefore prediction horizon. Short segments of at most 2–3 s ease the tension [30], and upgrading buffered tiles lets the fetcher revise a bet as the prediction sharpens. An edge cache moves the commitment closer to display, so caching and prediction are complements, not substitutes. Remote rendering faces the same horizon, with pose prediction hiding the round trip rather than throughput dips (Section 6).

A wrong bet is paid in one of two currencies: a thin margin shows low-quality or blank tiles, and a wide margin spends the saving. Tiling adds a fixed cost: a median coding overhead of 7.3–10.4% for 1-s tiles and a rise in Flare's phone CPU load from 18% to 58% [113], and up to 37.3% overhead for an 8×5 grid [46] (Section 4).

The net saving falls as the evaluation becomes real. Perfect knowledge of future motion bounds it at 80% [11]. HEVC tiles saved 65% with a static viewport and over 40% with recorded head motion [46]; Flare on phones over commercial LTE, with four users and three videos, saved 26–35% [113]. The gradient recurs in 6DoF: ViVo over commercial 5G saved 40% for headset users but only 10% for phone users watching the same content, because people move differently on each device [47].

It is established that viewport-adaptive delivery saves a substantial share of bytes; how large that share is on wireless headsets with real viewers, and whether viewers notice it, is not. Most evidence comes from one on-demand client on an emulated link, only two used commercial cellular traces [47, 137], and mainstream quality models failed on a database of 378 viewport-adaptive videos [89]. No user study has placed several headsets behind one bottleneck, the normal case at home or in a venue.

Open problems

- **P1. Do predictor rankings survive in closed loop?** Persistence nearly matches learned predictors offline, and accuracy is rarely tied to streaming outcome. Running extrapolators and learned predictors under one controller, over real bandwidth, with disjoint splits and a viewer rating study, would show whether accuracy gains reach viewers.
- **P2. Can calibrated uncertainty replace fixed margins?** Controllers need view probabilities but receive point estimates. Distribution-free calibration wrapped around any predictor, with margins chosen for a target miss rate, would show how much bandwidth honest uncertainty saves.
- **P3. What objective should the fetcher maximise?** Weights of quality, stalls and switches are chosen by hand, and quality models fail on viewport-adaptive streams. Fitting the objective to user ratings, or adopting tile extensions of standard models [69, 140], would make controller rankings meaningful.
- **P4. How should adaptation behave over real wireless access and shared bottlenecks?** Paired head-motion and radio traces from Wi-Fi and commercial 5G, replayed and run live with several headsets on one access point, would test whether radio hints and access-point coordination reduce stalls and unfairness.

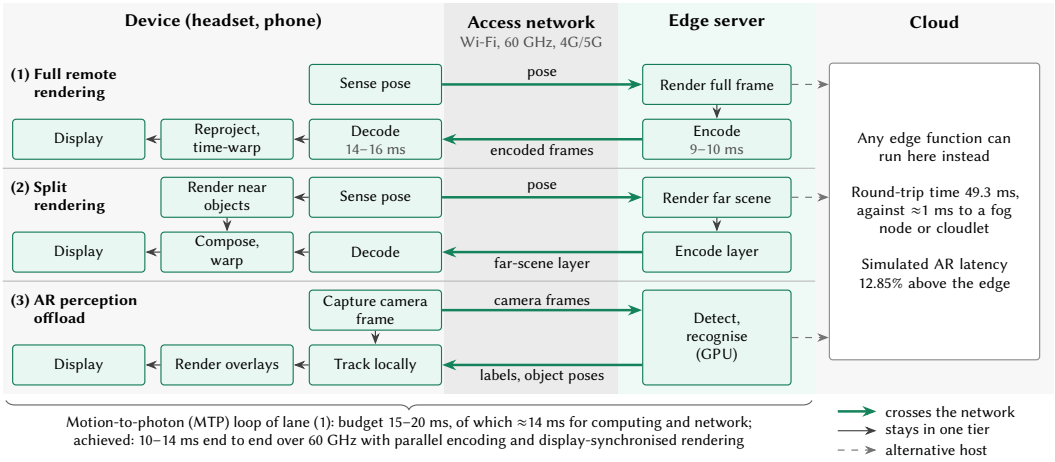


Fig. 4. Where each pipeline stage runs in full remote rendering, split rendering and AR perception offload. The split point is where the pipeline crosses the access network; display, and in AR also tracking, stay on the device in every design, which is what makes delay hiding possible. Values: MTP budget [36], 60 GHz end-to-end latency [86], encode and phone decode times [73], round-trip times [41], simulated edge advantage [116].

P5. What should an edge keep for 6DoF content, and do cache gains reach viewers? A level-of-detail cache shared by co-located users, and an HTTP edge serving tiled DASH under popularity, viewport-aware and Belady policies with viewers, would show whether hit-ratio gains are perceptible.

6 WHERE COMPUTATION HAPPENS: OFFLOADING, REMOTE RENDERING AND LATENCY

A headset has a few watts of power and a few milliseconds of slack. Offloading spends the second to save the first: it moves rendering or perception to an edge or cloud server and pays in network delay, which must still fit inside the deadline of the display loop, the motion-to-photon (MTP) latency. The central design variable is therefore the *split point*, the stage at which the sense–render–display pipeline leaves the device. It fixes what crosses the network, which direction carries the load, how stale the result is when it returns, and which corrections the device can still apply before display (Figure 4). This section follows four decisions: where to split, how to budget the delay a split adds, how to hide that delay, and what the split costs the device.

6.1 Where to split

The later the split, the less the device computes and the more bandwidth and staleness it inherits. Four families of design follow from this tension (Table 6).

Full remote rendering moves the whole renderer and streams finished frames; the device keeps only decoding, reprojection and display. Compression is mandatory: raw stereo frames at 2160×1200 and 90 Hz need 5.6 Gbps [86]. The design is simple and content-agnostic, but the codec sets a floor on delay.

Split or hybrid rendering divides the scene. Near or interactive objects are rendered on the device and the distant scene remotely; one such system used 20–30% of the bandwidth of full remote rendering and tolerated delays of up to 200 ms [91]. Others send an intermediate that the device can re-render from a new pose: colour plus depth for client-side warping [27], or one eye rendered

remotely and the other reprojected, which lowered MTP by 36% [64]. For neural scenes, streamed as pixels or as dynamic Gaussians [65], the right split is not yet known (Section 4).

Perception offload is the AR counterpart. The device streams camera frames upward, an edge GPU detects objects, and only labels and object poses return; tracking and overlay rendering stay local. The load now sits on the uplink. Once the server has a GPU, the network dominates inference latency, and sending frames as H.264 video rather than as images delivered about ten times more frames at 100 kbps [115].

Placement decides which server hosts the offloaded stages, and proximity is the one result that holds across methods. In a shipyard deployment, a fog gateway and a cloudlet answered in about 1 ms against 49.3 ms for a remote cloud, but the fog gateway was four times slower than the cloudlet at 240 clients, and the Wi-Fi hop was excluded by design [41]. Static placement breaks down once many users share an edge, because server queueing then dominates [165]. Adaptive controllers jointly optimise computation and communication, for example with deep reinforcement learning [26]. Their weakness is their inputs: a 2018 fit of detector accuracy against frame size [87] reappears in later formulations [172]. Learned placement is plausible but almost wholly untested on real render times and real links.

6.2 How to budget delay

A latency budget is a sum of stage delays, and it means something only when it names the loop it bounds. The common decomposition allows under 1 ms for sensing and about 5 ms for display within a 15–20 ms MTP bound, which leaves about 14 ms for computing and communication [36]. This is tight: a phone took 9–10 ms to encode and 14–16 ms to decode an H.264 frame [73], so decoding alone can exceed the network's share.

Pipelining and timing matter as much as link rate. Cutting the Cord split each eye's frame across four parallel encoders and aligned rendering with the client's display refresh. Over 802.11ad it reached 10–14 ms end to end against a sequential median above 26 ms [86]. In AR the uplink is the largest stage. In multi-user AR over LTE, median end-to-end latency was 12.5 s, the radio access network contributed 31.2%, and uplink bursts triggered TCP slow starts [8]; on 5G mmWave the uplink was still insufficient for baseline edge detection [45].

Two loops are routinely conflated. MTP runs from head motion to the updated image; interaction latency runs from a user's action to its visible consequence. In a crane simulator, display delays of 0–30 ms affected performance, whereas controller delay mattered only at 800 ms [16]. Quoted targets range from 5 ms [6] to 100 ms [24]. The spread is not noise: it reflects which loop is meant, and whether reprojection or local tracking already absorbs part of the delay.

Budgets are set on the mean, while users feel the tail. The 3GPP XR traffic model underestimated effective bitrate by 25% on one headset [71], so a budget met on average can still miss many frames. Latency studies seldom report percentiles or jitter, although a phone camera can measure delay to within 10 ms [40]. How jitter is perceived and how transport shapes it are treated in Sections 7 and 9.

6.3 How to hide delay

Every hiding technique lets the last stage on the device correct a result computed for an earlier state, so it can correct only what the device observes locally. Four families differ in what that is.

Reprojection (time-warp) shifts the last received frame to the newest head pose just before display. It is the established success. Post-render warping removed up to 80% of the aiming penalty of 80 ms added latency [67], and stalls of up to four frames at 90 Hz went unnoticed [66]. With reprojection active, ALVR users hardly noticed round-trip times up to 90 ms [83], although CloudXR quality fell

from 75 ms [119]. Its limit is fundamental: it corrects the viewpoint, not the scene. Moving objects, disoccluded regions and the outcome of a user's action remain stale.

Prediction and pre-rendering render for where the user will be. Rendering beyond the field of view, sized to the current latency, reduced delivery latency by about 22% in a simulation driven by real user data [96]. Such methods trade bandwidth for margin, and the prediction horizon bounds their gain (Section 5). *Speculation* goes further and pre-renders every reachable state: Furion prefetched panoramas for each grid point at about 132 Mbps [73]. It works only where the state space is small.

Local tracking hides the staleness of remote perception. Without it, 16.7 ms of offload latency raised false detections from 0 to 31.56%; with tracking and region-of-interest encoding, offload latency also fell from 34.56 to 17.23 ms on 2.4 GHz Wi-Fi [85]. CloudAR held 30 frames per second with at least 97% recognition accuracy [166]. Local tracking is the established design rule of edge AR.

Hybrid fallback keeps interactive objects on the device, so the delay-sensitive part never crosses the network [91]. No family hides loss: ALVR users rated packet loss as worse than added delay [83], so delay hiding offers no protection on an unreliable link. Nor can any family hide the state of other users (Section 8).

6.4 What it costs the device

Device energy is the main argument for offloading and its most contested result. Models promise savings of up to 85% [172], but measurements disagree. Edge offloading halved smartphone energy for AR on average [9], whereas a Quest Pro drained its battery 15% faster with wireless offloading than with local rendering [61], and Air Link on a Quest 2 also cost more energy [117]. DeepDecision measured about 2900 mW for offloading regardless of bitrate and frame rate [115], whereas most models scale energy with transmitted data. The models omit the radio's tail energy and the power of the decoder and display. Offloading moves load from the GPU to the radio and the decoder, so the saving depends on device, codec and link. None of these studies reports heat, so the thermal trade-off, which also bounds sustained local rendering, is unmeasured.

Open problems

- **P1. Which split point is best for a given scene, device and link?** Frame, layer, depth and Gaussian splits are each evaluated against their own baseline. A common benchmark with fixed scenes, devices and network traces would rank them.
- **P2. Do learned placement controllers survive real render times and real links?** They are trained on synthetic task sizes and channels. Replaying measured render times, head motion and 5G or Wi-Fi traces would show which gains hold.
- **P3. When does offloading save device energy?** Models predict large savings; headset measurements found the opposite [61, 117]. Component-level measurements of radio, decoder, display and GPU across headsets, codecs and links would settle the direction.
- **P4. How should a budget account for the tail and for interaction?** Budgets target mean MTP, while users feel late frames and delayed consequences of actions. Studies varying the delay distribution and the affected loop, with reprojection on and off, would yield derived budgets.
- **P5. Can edge AR scale on a shared uplink?** Real-network evidence is single-user and pessimistic [8, 45]. A 5G or Wi-Fi testbed scheduling several AR camera uplinks in one cell would test whether perception offload scales.

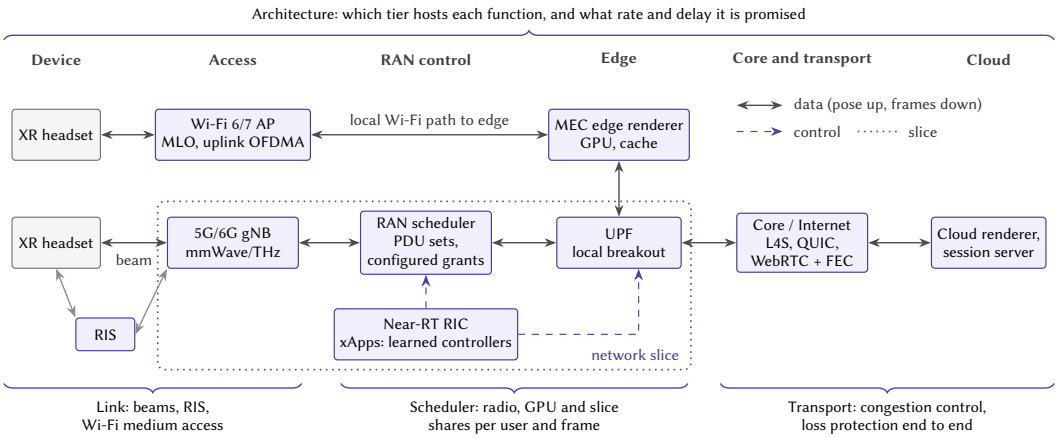


Fig. 5. End-to-end 5G/6G and Wi-Fi delivery chain for XR, with the layer at which each subsection makes its decisions. The link acts on the last hop, the scheduler inside the slice, transport across the whole path and architecture across all tiers; most designs optimise one layer and hold the others fixed.

7 PROVISIONING THE NETWORK: RADIO, RESOURCES, TRANSPORT AND ARCHITECTURE

XR traffic is unlike the traffic that networks were built for. It is *periodic*: the renderer emits a frame every 11–17 ms and the headset reports pose at least as often. It is *bursty*: each frame leaves the encoder as a block of hundreds of kilobytes that must cross the link before the next one [86]. It is *bidirectional*: pose, input and, for edge AR, camera frames go up while rendered content comes down. It is *deadline-bound*: a frame that misses its display slot is worth no more than a lost one. Best-effort networks assume elastic flows that value average rate; XR breaks that assumption. Provisioning must therefore supply rate, delay and reliability per frame, not per flow, and the tail of frame delivery, not its mean, decides whether the user sees a stall. Traffic volume depends on where computation is split (Section 6): a remote renderer sends video, a local renderer sends scene state. This section works bottom-up (Figure 5): link, scheduler, transport and architecture.

7.1 The link: a directional channel on a moving head

The last hop must carry multi-gigabit bursts to a head that turns and is easily blocked, and its tail, not its peak rate, is the limit. Four families address it.

Directional links (mmWave, THz) buy rate with narrow beams and pay in fragility. THz analysis gives 99.999% reliability at 20 ms with guaranteed line of sight, but with blockage reliability falls to 96% and tail delays reach 30–90 ms while the mean stays below 20 ms [21]. Commercial hardware lost up to 4.72% of packets under 45°/s head rotation [132].

Diversity and pose-steered beams are the remedy that holds. Dual-beam reception with coordinated access points raised reliability by up to 22.8% on real headset motion [50]. Beams steered from head pose reuse the motion predictors of Section 5, so prediction error bounds their gain.

Reconfigurable surfaces (RIS) promise to route around blockers, but the promise is assumed. RIS-assisted XR has never run on hardware; learned RIS control violated a 20 ms bound beyond 15 simulated users [88], and optimisers on practical prototypes take minutes [132] where they must react within a head turn.

Shared medium access binds on Wi-Fi and 5G before physical rate does. With frames aligned to the 802.11ad beacon interval, eight headsets at 400 Mbit/s met a 1 ms bound [133], but uplink OFDMA with three headsets disrupted tracking on real hardware [100]. On 5G-Advanced, code-block-group retransmission with outer-loop link adaptation reports up to 67% more satisfied XR users [108], in system-level simulation only.

7.2 The scheduler: deciding per frame who gets what

A scheduler divides spectrum, power, edge GPU time and cache among users at every slot. These resources substitute: cached fields of view and on-device computing can stand in for bandwidth [138]. The tension is between the model a scheduler optimises and the traffic it meets; Table 7 groups four families.

Optimisation wins when its models hold: multicasting overlapping 360-degree viewports over mmWave under a Lyapunov delay constraint raised HD delivery by 40–50% and cut 99th-percentile delay by 17–37% against unicast, in simulation driven by 50 real head traces [110]. *Heuristics* trade guarantees for deployability: a collaborative-VR allocator reached at least half the optimum and ran on commodity phones [23]. *Learning* schedulers beat greedy, random and proportional-fair baselines [25], but these baselines are weak, and learning and optimisation are rarely compared on the same traces. On an O-RAN testbed, latency-estimation control met 10–16.67 ms targets but over-provisioned, while packet-size prediction was efficient but lost latency control [19].

XR-aware mechanisms change what the radio sees rather than how it optimises. A PDU set lets the RAN drop or prioritise a frame's packets together under one delay budget; a delay-tracking PDU-set scheduler served 4–5 times more satisfied users than earlier PDU-set-aware schedulers under scarce resources [72]. Discontinuous reception aligned to the frame period saves headset power [80], and configured grants could pre-allocate the periodic pose uplink, but no testbed has compared them with dynamic grants.

Three problems sit above the family choice. *Joint radio–rendering allocation* treats GPU capacity as a free variable [35]; only one testbed adapted base-station bandwidth and edge GPU frequency together [105]. *Multicast* gains depend on viewport overlap [110], so they are unproven for interactive 6DoF content. *The objective* is contested: schedulers optimise a utility of tracking and transmission delay [25], an attention-weighted "Meta-Immersion" score [35] or weighted sums of quality and delay [23]. Each reports gains of 20–37% in its own metric, none validated with users (Section 9). Traffic inputs are optimistic too: the 3GPP baseline model underestimated effective VR bitrate by 25% against a measured headset [71].

7.3 Transport: congestion and loss for interactive frames

End to end, frames cross shared bottlenecks where loss-based congestion control fills queues, and a retransmission usually arrives after the frame's deadline. Users find loss the greater threat: in cloud VR, RTT up to 90 ms was hardly noticed while loss up to 8% mattered [83]. Stalls of up to four frames at 90 fps went unnoticed, and reprojection helped from eight frames upward [66], a margin that the compute pipeline sets (Section 6). Three families address the problem.

Low-queue congestion control keeps the bottleneck short. An L4S pipeline for volumetric conferencing converged 45% faster than Google congestion control and lost no frames where WebRTC lost 3% [42], but on a controlled bottleneck with one competing scheme; radio links with fast capacity swings are untested. *Session protocols* decide how media and control share a connection: Media over QUIC, RTP over QUIC and WebRTC have been compared for holographic conferencing over Wi-Fi and 5G [97], and aligning frame delivery with the display's VSync removed up to 14.3% of frame misses [86]. *Loss protection* repairs damage without waiting for a round trip, yet forward error correction for remote rendering has barely been studied [56], although users weigh loss most.

7.4 Architecture: where functions sit and what the network must promise

Architectures decide which tier hosts rendering, state and caching, and what the network must deliver. Four families recur. *Edge-centric* designs make edge computing a necessary tier [157]. *Sliced* designs isolate XR traffic and have testbed evidence: AR, cloud gaming and video calls were measured under RAN-slice configurations on an O-RAN 5G network [62]. *6G and metaverse visions* add THz access, AI-native control and ledgers [22], but rarely map their functions onto the 5G-Advanced XR mechanisms that already exist, so what 6G must add remains unstated. *Measured platforms* tell a plainer story: throughput, latency and device load on five social VR platforms grew almost linearly with users because servers forward avatar data directly [29]. The scaling bottleneck there is server logic, not access capacity (Section 8).

Requirements need the same discipline as results. Stated per-user rates exceed measured ones by two to five orders of magnitude (Table 8) because the operating point is left unstated: an uncompressed light field is not a compressed point cloud. A requirement means something only with its representation, codec, split point and control loop attached. The figures also propagate: one survey reproduces another's requirement table with a reliability target changed from $1 - 10^{-6}$ to $1 - 10^{-5}$ [5, 157], so the apparent consensus is partly one source counted several times. Ledgers, finally, are placed in the interactive data path for ownership and trust [22, 157], yet no study has measured the delay they add.

Open problems

- **P1. Which objective should an XR scheduler optimise?** Allocators optimise incompatible QoE constructs that no user study has validated. A QoE-in-the-loop slicing study on an open 5G testbed, comparing static, latency-target and QoE-model xApps with ratings and sickness scores, would decide which objective matters.
- **P2. Do joint radio–rendering allocators survive measured costs?** Rendering capacity is a free variable in these models. Profiling render and encode time and GPU power on edge hardware, then re-running published allocators under measured costs, would show whether their rankings hold.
- **P3. Can RIS or THz links meet frame deadlines for real XR traffic?** RIS work is simulated and THz reliability rests on Poisson traffic. A pose-driven RIS prototype at 28 or 60 GHz, measured on frame deadline misses under head motion, would test the premise.
- **P4. How should the uplink be granted?** Pose and camera uplink enter motion-to-photon latency directly, yet configured, dynamic and pose-prioritised grants have never been compared on a testbed with tracking error as the outcome.
- **P5. How should congestion control and loss protection share a frame deadline?** Users weigh loss above delay, but L4S and forward error correction are tested only on emulated bottlenecks. Comparing them over cellular and Wi-Fi with cross traffic, reporting 99th-percentile frame latency, would settle the design.
- **P6. What do shared worlds actually require?** Requirement tables are copied, not derived. Traffic measurements of shared-world platforms across user counts and avatar fidelities, compared with standard XR traffic models, would replace them.

8 CONNECTING PEOPLE AND THINGS: SHARED WORLDS, TOUCH AND TELEPRESENCE

Once several people or machines share one world, delivering media is not enough. The network must keep endpoints consistent and mutually responsive: a gesture must arrive in time to be answered, and a moved object must rest in the same place for everyone. Three kinds of loop run through such a session, and different things bound them (Figure 6 and Table 9). A control loop, such as bilateral haptic teleoperation, closes a feedback system across the network, so delay threatens

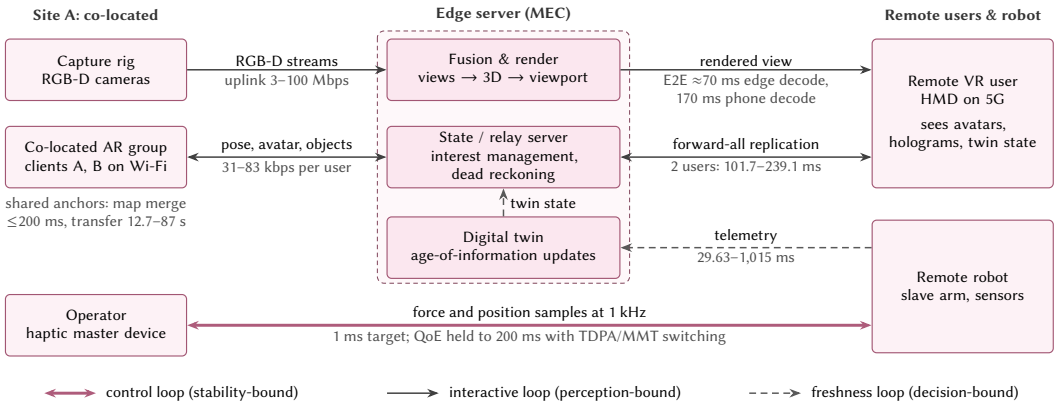


Fig. 6. Architecture of a multi-party XR session: a co-located group and a remote user share state through an edge relay, a capture rig feeds edge fusion for holoportation, and an operator drives a remote robot whose digital twin updates every viewer. The loops differ in what bounds them (line style), so no single latency target serves the session. Numbers are measured values cited in the text: avatar rates and replication latency [29], holoportation rates and latency [37, 126], anchors [34, 143], twin updates [20, 60], haptic rate and tolerance [2, 130, 159].

stability before it threatens experience. Interactive loops carry shared state and remote people; perception bounds them, because delay matters when users notice it or when it changes how they act together. Freshness loops keep a digital twin in step with a physical asset; the decision the viewer must take bounds them. The section is organised by what is shared: state, people, touch, and machines. Representations belong to Section 4, radio and transport to Section 7, and group experience to Section 9.

8.1 Shared state: who holds the truth, and who sees what

Every shared world must decide where the authoritative copy of its state lives and how much of it each client receives. Three families answer differently. *Authoritative relays* order all updates at a server and forward them. Five commercial social VR platforms forward every avatar update unprocessed, so throughput, latency and client load grow roughly linearly with users; two-user end-to-end latency was 101.7–239.1 ms [29]. A relay gives one ordering of events but sits in every path: routing through a cloud middleware raised round-trip time from 0.04 s to 0.6 s [170]. *Replicated, optimistic state* lets each client apply its own actions at once and reconcile later, as in decoupled edge physics serving over 100 users [70]. It buys responsiveness with conflicts: two users actuating one digital twin without a coordination policy conflicted in 33–56% of time windows [95]. *Peer-to-peer sharing* removes the relay; co-located multicast rebuilt scenes several times faster than unicast TCP for groups of 7 and 16 [173]. No study has compared the three families on the same workload, and which inconsistencies users notice is assumed rather than measured, although synchronisation faults were the most common symptom in 2,649 bug reports from multi-user XR software [82].

Scaling fails at replication before it fails at link capacity. In a controlled study, update intervals grew from 18 ms with 2 clients to 158 ms with 125, and perceived fluidity collapsed beyond 25 avatars [74]. Networked games use two remedies. Interest management sends each client only the state relevant to it; dead reckoning sends a motion model and updates it only when the receiver's extrapolation error exceeds a bound. XR synchronisation has barely adopted either, and neither has been tested with XR users.

Co-located AR adds a shared coordinate frame, which becomes the bottleneck. Edge map merging finished within 200 ms [34], yet anchor transfers took 12.7–87.18 s [143].

Fairness between users is the most contested question. Predictive buffering cut inter-client skew by 76% for 100 clients in a controlled testbed [125], but such schedulers assume that equal delay means equal experience. User evidence points elsewhere: dyads perceived network impairment yet completed tasks no slower [33]. The largest controlled study used 30 pairs [103], and a review of 56 studies found that almost all optimise latency for an individual, not a group [114]. Group quality of experience (QoE), treated in Section 9, is unmeasured beyond pairs.

8.2 People: a ladder of fidelity

A remote person can be sent as a few joint angles or as a live volumetric capture; each rung up buys realism with bits, latency and capture hardware. *Pose and parametric avatars* are cheapest: 216 B per update for a 17-joint skeleton [74], or 31–83 kbps per user on VRChat, Rec Room and Hubs, although Horizon Worlds exceeded 4.5 Mbps downstream at 15 users [29]. *Semantic, model-driven avatars* send features that drive a pre-built model of the person; Mon3tr animated a Gaussian avatar with under 0.2 Mbps at about 80 ms [84]. This family assumes an enrolled model and a stable feature stream; its robustness to loss is unsettled.

Volumetric telepresence captures, uplinks, fuses and renders live geometry, and each stage has its cost. Tiled user-adaptive streaming cut a person's raw point cloud from about 300 Mbps to 14 Mbps [135]. An operator trial streamed 5–100 Mbps, with end-to-end latency near 170 ms when phones decoded and 70 ms when the edge did [37]; depth packed into H.264 needed 3–30 Mbps [126]. At the top, *light-field booths* such as Project Starline need dedicated rigs [75].

Two assumptions go unexamined. Calls are treated as downlink problems, yet the holographic uplink has rarely been measured [97]. Semantic schemes report savings in PSNR, never against geometry at equal bitrate with users. Whether 3D beats 2D is contested: Project Starline [75] and stroke consultations with 30 outpatients [151] favour 3D, while a VR2Gather–Zoom comparison rated 3D below 2D video [127]. All held the network fixed.

8.3 Touch: closing a control loop across the network

Haptic teleoperation is the only loop here that closes a feedback system, so delay can destabilise it. The field set its requirement early at 1 ms end-to-end latency with 99.999% reliability [2]. Control theory offers three families that relax it. *Wave variables* encode force and velocity so that the channel stays passive, but they handle only constant delay. *Time-domain passivity* (TDPA) monitors the energy the network injects and dissipates it; with perceptual coding it stayed stable under time-varying delay of 100 ± 30 ms while sending up to 80% fewer packets. *Model-mediated teleoperation* (MMT) renders force from a local model of the remote environment and sends only model updates; it cut data by about 90% and stayed stable at round-trip delays up to 1 s [7]. MMT assumes an environment that changes slowly enough to be modelled; TDPA suits fast-changing contact at low delay [158].

Perceptual coding is orthogonal to these families. Deadband coders send a sample only when the change would be felt, using Weber fractions of 7–15% for force, and reduce packet rates by 80–90% [130]; codec details belong to Section 4. Co-designing receiver-side prediction with radio resources saved up to 77.80% bandwidth [68], but only open-loop on recorded trajectories, not with a live network, a stable controller and users.

Measured tolerance is far looser and depends on the controller. Switching between TDPA and MMT held QoE at delays up to 200 ms [159]. The 1 ms figure assumes a controller without delay compensation; it is not a measured limit of human touch. Haptic streams rarely travel with their

video, and cross-modal work questions their usual priority: deliberate haptic loss with visual restoration raised opinion scores by more than 80% at 30% loss [139].

8.4 Machines and places: keeping a twin fresh

A digital twin reverses telepresence: the network must keep a replica of a physical asset fresh enough for the viewer's decision. The natural metric is age of information (AoI), the time since the newest received update was generated. Two families use it differently. *Freshness optimisers* allocate radio and edge resources to minimise AoI or synchronisation time; optimal-transport association cut synchronisation time from 31.23–51.23 ms to 19.11–25.17 ms [48]. They assume their workloads, such as 10 Mb per update, and rarely leave simulation; a task-oriented design on a real robot arm, which halved the required packet rate, is an exception [98]. *Measured pipelines* report what deployed stacks deliver. A robot vehicle mirrored on a headset updated in 29.63 ms over a local network [60], whereas a cobot driven through a commercial IoT platform took 1,015 ms end to end [20]. Each figure was called adequate; none was set against a task requirement.

Sending state rather than pixels is a consistent lever: pose streaming kept remote work usable under added latency where video did not [162]. Once users act through the twin, the freshness loop becomes a shared-state problem, with the conflicts seen above [95]. Freshness metrics have multiplied, yet none has been tested against what a viewer perceives or decides.

Open problems

- **P1. What makes a group experience fair?** Skew schedulers assume equal delay means equal experience, but controlled studies stop at pairs [33, 103]. Groups of two to four under per-user impairments, scored on a shared task and perceived fairness, would give schedulers a validated target.
- **P2. Can interest management and dead reckoning scale social XR?** Platforms forward every update and scale linearly [29]. Area-of-interest filtering with error-bounded extrapolation on an open platform, with a perceptual study of avatar error, would show how far replication cost can fall.
- **P3. How much delay does touch tolerate in XR tasks?** The 1 ms target persists while users tolerated 200 ms with adaptive control [159]. Factorial delay, loss and jitter studies across TDPA, MMT and closed-loop predictors would yield tolerance curves for provisioning.
- **P4. How should haptic flows share a link with XR video?** Haptics and remote-rendered frames are scheduled apart. Joint scheduling on Wi-Fi or 5G, measuring haptic tail delay, motion-to-photon latency and inter-stream skew, would answer it.
- **P5. Which rung of the fidelity ladder serves a conversation?** Semantic avatars were never compared with geometry at equal bitrate, and 3D-versus-2D studies hold the network fixed. Two-way calls varying delay, loss and uplink rate against a matched 2D control are needed.
- **P6. How stale may a twin be?** AoI is optimised, yet no study links staleness to detection or control errors [20, 48]. Psychometric staleness curves for monitoring and actuation tasks would turn freshness budgets into requirements.

9 ASSURING EXPERIENCE AND TRUST: QOE, SECURITY AND PRIVACY

The previous sections optimise parts of the loop against targets. This section asks whether the result is good enough for its users, and safe for them. Experience and trust share three properties. Both belong to the whole loop: a frame arrives late because of the encoder, the scheduler, the transport pacing and the reprojection together, and a pose leaks whether it travels to a renderer, a cache or a peer. Both are judged at the human end, by what a user perceives and by what an adversary learns. And both are usually inferred from proxies: network metrics stand in for experience, and attack accuracy on a captured dataset for risk. The first half asks what users weigh and how experience

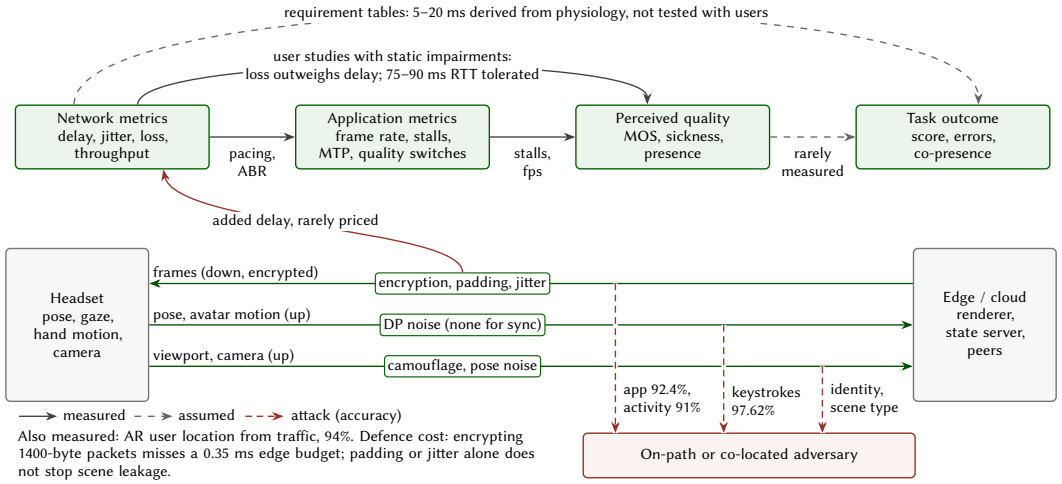


Fig. 7. Measurement chain and attack surfaces of a networked XR session. Solid arrows are links that user or traffic studies have measured, dashed arrows links that are assumed; red taps show measured attack accuracies, and defences sit on the flows where they add delay that has rarely been priced against the frame deadline.

and traffic are modelled; the second asks what XR traffic reveals and what defences cost. Figure 7 draws the measurement chain, the attack surfaces and the points where defences add delay.

9.1 What users weigh

Users of remote-rendered XR weigh loss and stalls above added delay of tens of milliseconds. On ALVR, round-trip times (RTT) up to 90 ms were hardly noticed, while 8% loss and a 35 Mbps cap degraded experience [83]. With CloudXR (N=30), QoE fell at RTT of 75 ms or more, at loss above 6%, or at 25 ms combined with 4% loss [119]. In social XR an RTT up to 77 ms had no significant effect, while jitter did [101].

The headset separates two latencies. Rotation latency, the motion-to-photon (MTP) delay of turning the head, is served locally by reprojection; network RTT reaches only interaction latency, the delay of aiming, grabbing or seeing a peer react (Section 6). Isolated stalls of up to four frames went unnoticed, and reprojection helped from eight frames (N=29) [66]. This explains a contested budget. Requirement papers derive 5–10 ms RTT and 20 ms interaction latency from physiology [55], four to ten times below what users tolerated. No study has varied RTT and latency compensation together, so the budget that edge placement should meet remains open.

The network moves perceived quality more than presence or sickness. Viewport-adaptive delay lowered perceived quality but barely changed presence and left sickness almost untouched (20 observers) [31]. Fewer than 22% of viewers reported sickness for AR point clouds, and two quality levels were indistinguishable at 5 m on a HoloLens 2 (N=32) [104]; content motion and display, not the network, seem to drive sickness. Task performance is rarely measured beside ratings. Group effects are tested almost only in pairs with the impairment on one link [14]; who is delayed in a larger group, and how the others judge it, is unknown (Section 8).

These results share one study design: *static impairments*. Every user study applies constant RTT, independent loss or throttling; real 4G and 5G measurements enter only afterwards, as inputs to models fitted under constant impairments [118]. No study has *replayed measured network dynamics*

to users. Wireless loss is bursty and correlated, so thresholds learned under independent loss may not transfer.

9.2 How experience and traffic are modelled

A QoE model maps what can be observed to what a user would report. Four families differ in what they observe and what they were validated against (Table 10). *Parametric standard models* extend ITU-T P.1203 and P.1204.3 to tiled streaming [140], with P.919 as the test method for short stimuli. *Objective media metrics* are established for coding impairments: VMAF correlated best on a public dataset [43], and colour PSNR reached $r = 0.84$ for point clouds, depending on content [104]. Neither family sees interaction latency. *Learned estimators* add it but fit locally: a cloud VR gaming model reached a Pearson correlation of 0.93 [77]. *Network-side estimators* infer quality from what an operator sees, such as encrypted 360-degree traffic from drive tests [59], and are validated against formulas, not people. PERCEIVE predicted a parametric QoE index within 4%, but that index was never checked against ratings [32].

All four families lack *temporal pooling*. P.919 uses 10 s stimuli, while presence needs multi-minute tasks [109], and no model pools sub-second network events into the QoE of a session.

Measurements of commercial platforms show that the application stack, not the link, often sets what users see. WebRTC pacing in 5.56 ms batches set frame delay at 6–21 ms over Wi-Fi 6 with an RTT of 1.5 ms [18]. Under 27 Mbps throttling, adaptive bitrate kept cloud VR frame loss at or below 5.12% where a fixed 40 Mbps stream lost 32.51% [169]: adaptation converts a throughput shortfall into image quality rather than lost frames. Horizon Workrooms used 2–3 Mbps downlink and 0.6 Mbps uplink [28], two orders of magnitude below the per-user rates in metaverse visions (Section 7). *Statistical traffic models* capture constant-bitrate video well, but they are open loop. The open RiftCat model matched mean but not 95th-percentile delay [76], and the 3GPP baseline underestimated effective VR bitrate by 25% [71]. None couples user motion with rate adaptation and congestion control, so schedulers evaluated on them misjudge the tail.

9.3 What XR traffic reveals

XR transmits the body. Pose, gaze, viewport requests and camera frames are control signals and, at the same time, behavioural and biometric data. Encryption hides content but not size, timing or recipients.

Traffic metadata reveals place and activity, an established result. Traffic patterns of a mobile AR application located users with 94% accuracy [99]. On Meta Quest Pro, off-the-shelf classifiers identified applications with 92.4% accuracy and activities with 91%, from under 10 minutes of traffic each [102]. *Motion as a biometric* is the sharpest channel. Avatar motion received by an adversary's own client in multi-user VR yielded 97.62% of typed keystrokes across four applications [134], so state synchronisation, which broadcasts head and hand poses to every peer, leaks by design (Section 8). *Viewport requests* identify users because prediction-driven tile requests follow individual head motion (Section 5); the leakage persists under federated prediction once prediction errors or QoE are uploaded [156]. *The camera uplink* of offloading AR glasses reveals the type of physical scene through packet sizes and timing [53], so the split chosen in Section 6 is also a privacy decision.

9.4 Defences and their price

Every defence spends a resource the loop is short of: delay, bandwidth or prediction accuracy. The price is rarely measured.

Encryption is cheap for small packets and expensive at the tightest budgets: encrypting 1400-byte packets did not fit a 0.35 ms edge budget, while 256-byte packets did [154]. *Obfuscation* hides

metadata at a cost in bandwidth or quality. Padding or jitter alone did not stop scene leakage, and combined defences conflicted with QoS [53]. Camouflaged tile requests need more resources and degrade prediction [150]. Noise remapping reduced the QoE loss of privacy noise by 55–100%, to 0–3% at maximum privacy [149], but that loss is computed from prediction accuracy; no user has been asked.

Authentication schemes for metaverse avatars are proven formally and costed in operation counts [120]; none is measured during an edge or platform hand-off, where re-authentication falls on the interaction path (Section 7). *Detection* rarely sees XR traffic: "metaverse" intrusion detectors report 96–99% accuracy on IoT and enterprise datasets that contain no XR flows [44]. *Blockchain* is contested: a 10-user pilot kept 88 FPS at 142 transactions per second [112], while most ledger frameworks report no latency at all. The question is whether a ledger write sits on the interaction path.

The privacy–QoE trade-off is therefore modelled, not perceived. No testbed study reports a defence's latency together with QoE or frame rate. A defence should be reported as a point on a privacy–QoE plane: attacker accuracy against added 99th-percentile frame latency and bandwidth, on a real split-rendering pipeline.

Open problems

- **P1. Do network-side estimators predict what XR users report?** Estimators target parametric indices, and none addresses interactive remote rendering. A public benchmark pairing encrypted packet traces, client frame logs and per-minute ratings for cloud VR would let them be compared against people.
- **P2. Do thresholds from static impairments hold under real wireless dynamics?** Users have only seen constant RTT and independent loss. Replaying measured Wi-Fi and 5G traces, with reprojection on and off, would fix the latency budget for remote-rendered XR.
- **P3. How do sub-second events pool into session and group QoE?** No model links bursts of stalls to a multi-minute session, and groups beyond pairs are untested. Scripted burst schedules and asymmetric impairments in groups of four to six would answer both.
- **P4. Can a closed-loop traffic model reproduce the tail?** Open-loop models err at the 95th percentile. A generator coupling pose, rate adaptation and pacing, validated on several real headsets, would make scheduler studies credible.
- **P5. What does a defence cost against the frame deadline?** Measuring TLS, DTLS and QUIC with padding and jitter on split rendering, as attacker accuracy against 99th-percentile frame latency, would give providers a price; the same holds for re-authentication during hand-off.
- **P6. Can state synchronisation share avatars without sharing biometrics?** Broadcast poses reveal keystrokes. A protocol that perturbs or coarsens remote poses, tested for keystroke inference, co-presence and sickness, would close the channel.

10 CROSS-CUTTING GAPS

Read side by side, the open problems of the six functions repeat themselves in different vocabularies. A viewport predictor never tested in a closed loop, an offloading controller that never left the simulator and an allocation scheme whose QoE objective was never shown to a user are three instances of one shortfall. We condense the open problems of Sections 4 to 9 into nine themes (Table 11) in three families: the *evidence* that claims rest on, the *objectives* that designs optimise, and the *blind spots* that most designs leave out.

10.1 Evidence

The first family concerns what claims are tested against. *Shared data are missing* (T1). There are no public traffic traces for AR offloading, avatar streams, twin-state updates, holographic calls or haptic flows. No 360-degree dataset records head motion, network conditions and ratings in the same session, and no latency benchmark couples motion, network delay and photodiode-measured MTP. Even simple baselines are often absent. A persistence predictor reached a hit rate of 0.8355 against 0.8402 for a sequence-to-sequence model on the same data [79]. *Evaluation stops at the simulator* (T2). Whole technology lines, among them reconfigurable surfaces carrying XR, terahertz XR links and metaverse resource markets, have never met hardware. Where learned controllers were tested on unseen users or content, they degraded [148, 152]. *Standards are not the substrate* (T3). Standard XR traffic models appear almost only in radio simulations. When one was checked against measurement, it underestimated cloud-VR bitrate by 25% [71]. The three gaps share a cause. Traces and testbeds are costly to build, and the venues that publish algorithms rarely credit them.

10.2 Objectives

The second family concerns what is optimised. *Network metrics are not linked to experience* (T4). Most objectives called "QoE" are formulas whose weights were chosen rather than fitted. Where users were asked, their answers diverged from the proxies. Packet loss outweighed delay in VR cloud gaming [147]. Users of collaborative VR could not tell forms of latency apart [33]. CloudVR users did not notice 20 ms of added latency [63], while offloading designs optimise to budgets of 5–13 ms. *Requirements are inherited* (T5). Visions state 100–250 Mbps per metaverse user [81, 157], while a measured commercial social-VR platform used 2–3 Mbps downstream [28]. Holographic figures range from 5–100 Mbps for compressed calls [37] to 6.63 TBps [3]. Haptic users tolerated up to 200 ms with adaptive control [159], against the 1 ms figure that motivates much tactile-internet work. A requirement that names neither the representation, the codec, the loop nor the percentile cannot be checked.

10.3 Blind spots

The third family concerns what is left out. XR sends a continuous upstream, yet the *uplink* (T6) is studied far less than the downlink, and where it has been measured it is the bottleneck. On public LTE, multi-user AR had a median end-to-end latency of 12.5 s, with the radio access network contributing 31.2% [8]. With three headsets on Wi-Fi 6, uplink OFDMA delayed tracking data [100]. *Scale* (T7) is untested. Measured social platforms scale linearly because they forward all avatar data [29], and controlled QoE studies use pairs of users. *Security is not costed* (T8). Attacks are measured and effective: keystrokes were inferred from avatar motion with 97.62% accuracy [134]. Defences, however, are seldom priced in frame latency. *Transport* (T9) has not followed the move from buffered streaming to interactive delivery. Users rank loss above delay [83, 147], yet forward error correction and L4S are rarely applied. An L4S pipeline for volumetric conferencing converged 45% faster than Google Congestion Control [42], but only in a controlled bottleneck.

10.4 How the gaps depend on each other

The gaps are not independent (Figure 8). Two are roots. Without open traces (T1), simulations cannot be validated (T2), requirements cannot be derived (T5), standard traffic models cannot be checked (T3), and uplink and multi-user workloads cannot even be described (T6, T7). Without a validated mapping from network metrics to experience (T4), results on allocation, transport and security (T8, T9) cannot be compared in units that matter to users. Two smaller questions point the same way. Device-energy savings from offloading rest on models, and the few measurements

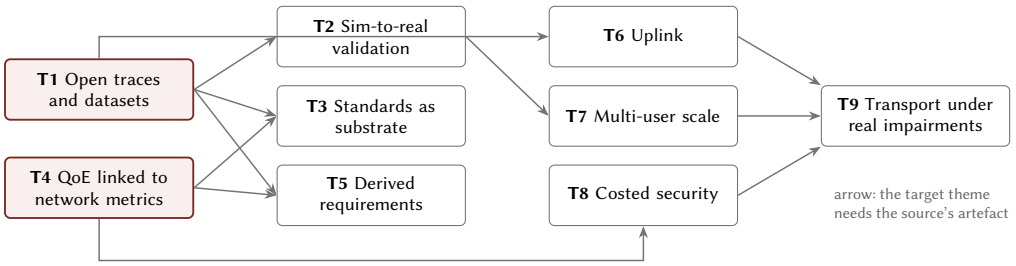


Fig. 8. Dependencies among the nine themes. Open traces (T1) and a validated mapping from network metrics to experience (T4) are the roots. Without them, simulations cannot be validated, requirements cannot be derived, and allocation, transport and security results cannot be compared in units that matter to users.

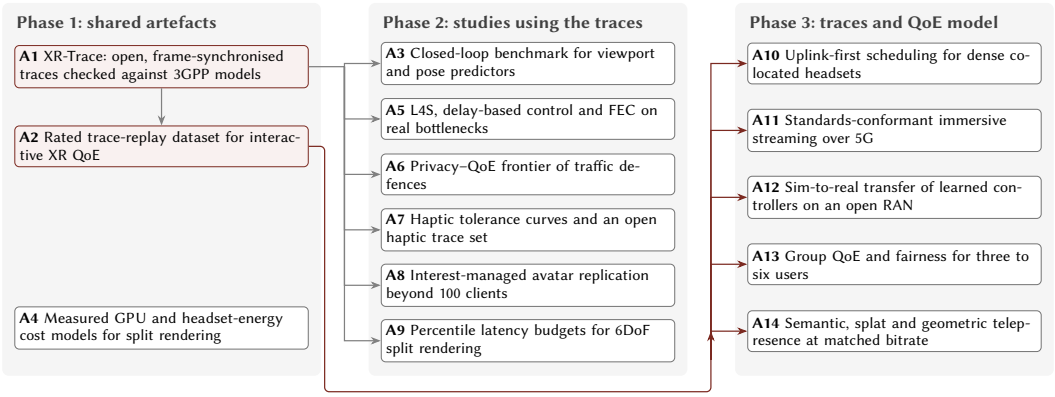


Fig. 9. Research agenda as a dependency roadmap. Phase-1 projects (red) build the artefacts that the others consume. A1 supplies traces to every later project (grey arrows); A2, built on A1, adds the validated QoE model that the Phase-3 projects optimise against (red arrows). A4 stands alone.

disagree: one standalone headset drained its battery 15% faster when offloading wirelessly [61], while an edge VR testbed found lower client power [106]. Neural scene representations have no tested rate adaptation, cache or split point. Both questions need the same two roots, and this ordering shapes the agenda that follows.

11 RESEARCH AGENDA

The gaps of Section 10 suggest fourteen projects that a networking or multimedia laboratory could start with existing hardware and open software. We order them by dependency (Figure 9). Projects that produce shared artefacts come first, because the others need them.

Phase 1: build what others need. XR-Trace (A1) would instrument the main workloads on 5G standalone and Wi-Fi 6E/7, with one, four and eight users. These are split-rendered VR, AR camera offloading, avatar and shared-state traffic, tiled 360-degree video and twin-state updates. Each trace would record pose, encoder output, packets at both ends, application events and photodiode MTP on a common clock. The project would then fit traffic models, compare them with the 3GPP XR models, and release the traces with a replay harness for ns-3 and NetEm. It comes first because every later project consumes it. The *rated replay dataset* (A2) would have participants perform interactive tasks while the network replays A1's traces. It would collect momentary and session

ratings, sickness scores, task performance and packet logs together, so that a QoE model can be fitted and then used as the objective by other work. It would also measure the difference between replayed dynamics and independent impairments of the same mean. *Energy cost models* (A4) would replace the assumed GPU and battery costs inside offloading optimisers with measured ones.

Phase 2 and 3: answer the open questions on shared ground. With traces in hand, five studies become cheap and comparable. They are a closed-loop predictor benchmark with persistence and Kalman baselines (A3), transport on real bottlenecks (A5), a privacy–QoE frontier for padding and jitter defences (A6), haptic tolerance curves (A7) and interest-managed replication beyond one hundred clients (A8). A percentile latency budget for 6DoF split rendering (A9) completes the phase. With a validated QoE model as well, the remaining projects can optimise against experience rather than a proxy. These are uplink-first scheduling (A10), a standards-conformant 5G pipeline (A11), sim-to-real transfer of learned controllers as xApps on an open RAN (A12), group QoE and fairness (A13), and a matched-bitrate comparison of semantic, Gaussian-splat and geometric telepresence (A14).

The evaluation bar. Each project should meet the standard that Transactions-level reviewers apply, and that much current work does not yet meet. First, a *baseline ladder*: a trivial baseline, a classical method, the strongest recent method re-implemented under the same conditions, an ablation and, where computable, an oracle. Second, *realism with control*: real hardware for the claim and trace replay for repeatability, with the gap between simulation and measurement reported. Third, *out-of-distribution tests* for anything learned, on held-out users, content and networks. Fourth, *sound statistics*: repeated runs with confidence intervals and, for user studies, a power analysis, within-subjects designs and mixed-effects models. Fifth, *reported cost and released artefacts*: run time, energy and signalling overhead on the target device, with code, traces and analysis scripts published.

12 CONCLUSION

Networked XR puts the network inside a perceptual loop. That single fact explains why its problems differ from those of video streaming or cloud gaming. Content must be chosen before the user looks at it. Rendering can move off the headset only if its delay can be hidden. Radio and transport must deliver frames, not flows, on time. Several people and machines must agree on one world. And quality must be judged at the human end, where both experience and privacy are decided.

Organising the field by function makes its structure visible. Six design questions cover it: what to encode, what to anticipate, where to compute, how to provision, how to connect and how to assure. Each has a small number of families of answers. Two couplings tie the functions together. The first is a shared deadline that every function spends from. The second is a trade between network and device that almost every design shifts one way or the other. Results rarely transfer between settings because they were obtained under a particular budget and a particular division of cost.

The weakest links are not algorithms but shared ground. The field lacks open, frame-synchronised traces of real XR workloads, and a model of experience fitted to people under realistic network dynamics. With these two artefacts, the questions that recur in every section become answerable on common terms: whether a learned scheduler beats a heuristic on a real cell, how much latency a defence costs, and what a hologram really needs. Without them, networked XR will keep producing designs faster than it can tell which of them work.

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Table 4. Representation families and where their cost lands. Figures are examples measured on the cited systems under different conditions; they locate each cost but are not matched operating points.

Family	Idea	Cost on wire	Cost at sender	Cost at receiver	Maturity
Image	360-degree video: sphere projected to rectangles, bits on the viewport	6–22 Mbps	low; tiles lose 7–10% efficiency	a decoder per tile, CPU 18→58%	standard (OMAF), deployed
Image	Multi-view plus depth (MIV): camera views, new views synthesised	30–100 Mbps	low; hardware video encoders	several decoders and view synthesis	standard; rarely tested on networks
Geometry	Point clouds (V-PCC, G-PCC, Draco)	19 Mbps live; raw up to 5 Gbps	V-PCC hours per second of content; G-PCC, Draco fast	3D rebuild; 30 fps on phones needs inter-frame coding	standard; on-demand prototypes
Geometry	Meshes: surfaces from fused depth	14 Mbps mean (reconstructed scene)	reconstruction	rasterisation	static codecs benchmarked
Learned	Neural scene (NeRF, 3DGS): model parameters per frame	0.53 MB per frame	training ≈1 min per frame	splat rendering, 37 ms per frame	research prototypes
Learned	Semantic and generative: meaning, not pixels	93–99.9% less	inference, shared knowledge base	generative inference; 13.7 s end to end	simulation, one 5G prototype
Signal	Haptic: samples sent only above a perceptual threshold	80–90% fewer packets	negligible, delay-free	hold or predict missing samples	standard (IEEE 1918.1.1)
Signal	Holographic: light field or phase pattern	6.63 TBps by arithmetic; 173 Mbps measured	V-PCC 11–42 min per second	display not characterised	requirement tables

Table 5. Families of approaches for the three anticipation decisions. Each family wins under its own assumption; none dominates once that assumption fails.

Decision	Family	Idea	Assumes	Wins when	Limit
Predict	Extrapolators	Persistence, linear or Kalman continuation of pose	Smooth motion over the horizon	Horizon <0.5 s; no training data	Blind to content and to sudden turns
	Learned sequence	LSTM or transformer over past trajectory	Training users and videos resemble test ones	Mid horizons on familiar content	Small margin over persistence; leakage-prone splits
	Content-aware, cross-user	Saliency, gaze or earlier viewers as priors	Content and viewers are shared	On-demand video at 2–10 s horizons	No prior for live or interactive XR
Fetch	Heuristic, buffer-based	Rules map buffer level and view probability to quality	Buffer reflects throughput	Unknown or shifting networks	Hand-tuned; no notion of optimum
	Optimisation	Rate-utility allocation over tiles, cells or LoD	Accurate throughput and view probabilities	Good estimates; 6DoF visibility known	Fragile when estimates are wrong
	Learned (RL)	Policy learned from trace replay	Deployment matches training traces	Stable, well-sampled conditions	Degrades on unseen users and networks
Keep	Popularity cache	LRU or LFU on independent tiles	Requests repeat as whole objects	Large caches; standard HTTP edges	Ignores viewport overlap
	Viewport-aware cache, prefetch	Place tiles or objects by predicted viewing	Viewers' attention overlaps	Popular on-demand content	Gains shown only against popularity baselines
	Shared delivery	Multicast or coded caching of common parts	Synchronised, similar requests	Co-located viewers of one stream	Asynchrony and subpacketisation

Table 6. Families of split point. Each family is defined by what crosses the network; that choice fixes which link carries the load, which delay the device can still hide, and where the design fails.

Family	What crosses the network	Uplink	Downlink	Hiding available on device	Typical limit
Full remote rendering	Encoded stereo frames	Pose and input, light	Heavy: compressed video (5.6 Gbps raw)	Rotational and translational reprojection	Codec delay; interaction stays stale
Split or hybrid rendering	Far-scene layer, one eye, or a warpable intermediate (depth, atlas, Gaussians)	Pose and input, light	Medium: a fraction of full frames	Local rendering of near objects; 6DoF warping	Device GPU load; disocclusion and composition artefacts
Perception offload (AR)	Camera frames up; labels and object poses down	Heavy: video frames	Light: results	Local tracking between detections	Uplink capacity; stale or wrong overlays
Placement across tiers	Any of the above, to fog, edge or cloud	As hosted family	As hosted family	As hosted family	Queueing at shared servers; untested learned controllers

Table 7. Families of XR schedulers. Each row names what the scheduler decides, what it takes as given, and where it wins and fails.

Family	Decides	Assumes	Strength	Limit
Optimisation (convex, Lyapunov, ADMM) [110, 138]	Cache, render and radio shares per slot	Known cost, QoE and channel models	Bounds; near-optimal at low run time	Breaks under model mismatch
Heuristics (greedy, approximation) [23]	Per-user quality and share	Monotone utility	Simple; provable fraction of optimum	No delay guarantee under bursts
Learning (DRL, bandits, xApps) [19, 25]	Admission, resource blocks, slice size	Test traffic resembles training traffic	Adapts to unmodelled dynamics	Weak baselines; generalisation untested
XR-aware mechanisms (PDU sets, XR DRX, CBG-HARQ) [72, 80, 108]	Frame delay budget, sleep, retransmission	Standard XR traffic model	Frame semantics in the RAN; deployable	System-level simulation only

Table 8. Stated requirements against measured or derived figures. The gaps follow from unstated content representation, codec and control loop.

Quantity	Stated	Measured or derived
Per-user metaverse rate	100–250 Mbps [81, 157]	Horizon Workrooms 2–3 Mbps down, 0.6 Mbps up [28]
Holographic call	1–10 Tbps [3]	5–100 Mbps for a compressed point cloud over live 5G [37]
Untethered VR link	15.75–26.25 Gbps uncompressed [132]	400 Mbit/s per 4K headset, eight per 802.11ad AP [133]
Interaction latency	1 ms for tactile use [123]	Haptic users tolerated 200 ms with adaptive control [159]; cloud VR RTT up to 90 ms hardly noticed [83]
Discomfort threshold	30 ms [81] vs <20 ms [123]	Not derived from a user study in either source

Table 9. The three loop types of a shared XR session. Measured tolerances come from the studies cited in the text; they show that each loop fails in a different way and needs its own network target.

Loop	What it carries	Bound	Delay measured or tolerated	Failure mode
Control (haptic teleoperation)	Force and position samples at 1 kHz	Stability of the bilateral controller	1 ms target; stable to 200 ms with TDPA/MMT switching [159], model-mediated to 1 s round trip [7]	Oscillation or unsafe contact force
Interactive: shared state	Pose, avatar and object updates, tens of kbps per user	Perception of consistency and co-presence	101.7–239.1 ms on deployed platforms [29]; impairment noticed, task time unchanged [33]	Divergent views, conflicting edits, unfair interaction
Interactive: remote people	Avatar parameters to volumetric video, 0.2–100 Mbps	Perception of the person	70–170 ms end to end [37]; avatar latency noticed at about 320 ms [52]	Degraded, frozen or vanished person
Freshness (digital twin)	Telemetry and state of a physical asset	The viewer's decision	29.63–1,015 ms, all reported adequate [20, 60]	Decisions taken on stale state

Table 10. Families of QoE models for networked XR. Each family is limited by what it observes and by the ground truth it was validated against.

Family	Observes	Validated against	Limit
Parametric standard (P.1203, P.1204.3 tiled)	Bitrate, resolution, stalls, quality switches	Ratings of short streaming stimuli	No interaction latency; 10 s stimuli
Objective media metrics (PSNR, VMAF, colour PSNR)	Decoded pixels or points against a reference	Ratings under coding impairments	Blind to timing; content dependent
Learned estimators	Network and application metrics, game state	Ratings on a few contents in one lab	Fitted and tested on the same small set
Network-side estimators	Encrypted traffic, radio and in-network counters	Parametric indices or activity labels	Never checked against users for interactive XR
Requirement tables	Human perception limits	Physiology, not networked user studies	4–10× stricter than tolerated RTT

Table 11. Nine cross-cutting gaps in networked XR, grouped into three families. Each recurs in at least three of the six functions.

	Gap	What goes wrong	Functions affected
Evidence			
T1	No open traces or benchmarks	Each design is tested on its own stack and workload; results cannot be compared or reproduced	all six
T2	Evaluation stops at the simulator	Learned and optimised controllers meet synthetic channels, traffic and motion, rarely a real link	anticipate, compute, provision
T3	Standards unused as baselines	Standard traffic models, codecs and QoE models are seldom the reference configuration, and seldom checked	encode, provision, assure
Objectives			
T4	Metrics not linked to experience	"QoE" objectives use chosen weights; user studies apply static impairments, not real dynamics	all six
T5	Requirements inherited, not derived	Rate and latency targets are copied between surveys without naming content, codec, loop or percentile	encode, provision, connect
Blind spots			
T6	The uplink	Pose, camera and capture streams are modelled as small or ignored; measured uplinks are the bottleneck	compute, provision, connect
T7	Multi-user scale	Systems and studies use one to a few users; group effects and fairness are untested	anticipate, connect, assure
T8	Uncosted security	Attacks are measured; defences are not priced in frame latency or QoE	anticipate, connect, assure
T9	Transport under real impairments	Interactive frames inherit assumptions from buffered streaming; loss resilience is rare	compute, provision, connect